CINZIA

AI / Machine Learning Engineer – Applied AI

Paris, France

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PROFESSIONAL EXPERIENCE

AI Systems Engineer — Framatome, DTI

La Défense, France

Primary project: End-to-end development of an industrial computer vision system for real-time zirconium defect detection

- **End-to-end AI development:** Contributed to the development of an industrial computer vision solution, from data preparation and analysis to model training, evaluation and deployment for real-time zirconium defect detection.
- **Data preparation & analysis:** Prepared and analyzed image datasets, investigated data quality and challenging samples, and contributed to preprocessing and data augmentation strategies to improve model robustness.
- **Deep Learning & Computer Vision:** Developed and optimized deep learning models for object detection, experimenting with different architectures and training strategies to improve detection performance in industrial conditions.
- **Model training & optimization:** Implemented and monitored training workflows in Python, analyzed model behavior and iteratively optimized architectures, hyperparameters and training strategies based on experimental results.
- **Model evaluation & validation:** Implemented evaluation workflows, monitored relevant performance metrics and analyzed false positives, false negatives and failure cases to assess model reliability and guide improvements.
- **Deployment & industrialization:** Contributed to the integration and deployment of AI models in an embedded industrial environment, considering inference latency, computational constraints and real-time processing requirements.
- **AI Engineering:** Developed Python-based workflows for model training, experimentation and inference, using reproducible development environments and engineering practices to support the transition from prototypes to operational AI solutions.
- **R&D & Generative AI:** Conducted R&D activities on emerging AI approaches, including Generative AI, Transformers and Large Language Models, investigating their potential applications to industrial use cases.

Technical Environment : Python, PyTorch, YOLO, Computer Vision, Deep Learning, Diffusion Models, Transformers, LLMs, JAX, Docker, Linux

EDUCATION

Double Master's Degree – Statistical Engineering & Data Science
ISUP & Sorbonne University

Paris, France | 2022 – 2024

- **Key coursework :** Machine Learning & Deep Learning, Advanced Statistics, Probability & Optimization, Computer Vision
- **Selected Academic Projects :** Time Series Forecasting (ARIMA, XGBoost, LSTM) · Online Convex Optimization & SVRG · Generative Models — Research Analysis

PROFILE

Machine Learning Engineer focused on building and industrializing robust AI systems from experimentation to production.

Strong background in applied mathematics, optimization, statistical modeling, and deep learning, with hands-on experience in computer vision, generative AI, and real-time ML systems.

Experienced in translating complex technical and operational constraints into scalable, reliable AI solutions, with a strong focus on measurable performance and real-world impact.

Interested in high-impact AI applications across healthcare, security, finance, and other decision systems.

SKILLS

- **Programming & Data**

Python · C++ · SQL · Bash

- **Machine Learning**

PyTorch · Computer Vision · Deep Learning · Statistical Learning · Time Series · Representation Learning

- **Generative & Foundation Models**

LLMs · Diffusion Models · Vision Transformers · Fine-tuning · Embeddings

- **Cloud & MLOps**

Azure · GCP · MLflow · Docker · Git · Linux · CI/CD · CUDA · Model Deployment · JAX

LANGUAGES

- French: Native
- English: C2
- Spanish: B2

Research Interests

- Foundation Models & Generative AI
- Model Reasoning & Interpretability
- Robustness & Evaluation
- AI Agents